

Co-design of virtual reality exergames for balance rehabilitation

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Figure 1: Balance rehabilitation exergame virtual environment (left). Therapists' preliminary tests (right).

ABSTRACT

Balance disorders, particularly in the elderly population, pose a significant public health challenge due to the associated risk of falls. Exercise-based therapies are effective for reducing this risk but face limitations like cost, accessibility, and sustaining patient engagement. By enabling personalized rehabilitation through engaging exergames, Virtual Reality (VR) offers a promising, cost-effective solution. This paper presents the collaborative design and preliminary evaluation of two VR-based games targeting specific balance skills. Developed in collaboration with a physiotherapist with extensive experience with patients with balance disorders, the games align with a rehabilitation protocol used by the therapist that categorizes training based on individual balance skills. Inspired by traditional physiotherapy exercises, the gameplay aims to train the same skills. Six health professionals agreed on the games' therapeutic potential, while recommending improvements in usability, difficulty adjustment and monitoring of patient progress.

Index Terms: Human balance, exercise games, virtual reality, rehabilitation.

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1 INTRODUCTION

Balance disorders and dizziness are prevalent worldwide, with a systematic review reporting lifetime prevalence of significant dizziness ranging between 17% and 30 % and for vertigo between 3% and 10% [22]. Moreover, data from the detailed National Health and Nutrition Examination Survey (NHANES) carried out in the US indicates that the proportion of balance disorders increases with age: with 18.2% of adults aged 40 to 49 years being affected while for the 50 to 64 years and 65 years or older groups the rate increases to 33.5% and 61.9% respectively [7]. Such disorders result in significant public health risk with balance impairments playing a key role in the risk of falls [29], which are a leading cause of injury-related hospitalization and a significant contributor to accidental deaths in the elderly population.

Fortunately, interventions for preventing falls in older people including exercise-based therapies that target balance, gait, and strength can reduce this risk [10]. A systematic review of such exercises found that programs that included balance training had the greatest effect on reducing falls [26]. Sustaining these programs requires involving professional teams of physiotherapists working in the public or private sector, posing a substantial cost issue for the patients and the healthcare system. In this clinical context, alternatives to a continuous care intervention from the physiotherapist during rehabilitation are viable as soon as individuals with a balance disorder actively collaborate in attending their own rehabilitation program, also at home, provided they are first carefully instructed and then regularly evaluated to assess the advance of the rehabilitation. As the success of a physiotherapeutic program largely depends on the attention and commitment with which the prescribed exercises are repeated, methods capable of engaging patients while rewarding their physical effort can avoid continuous monitoring by the

physiotherapist, in parallel reporting about the number, duration, and success rate of each rehabilitation session. Gamification of the exercises, or exergames [24], fits this goal: Interactive exergames have proven useful for improving balance in older adults [15], revealing the activation of interactive functions in patients yet, at the same time, calling for the inclusion of a more informative progress reporting during the game [11].

The intersection between computer gaming and balance tasks became clear early with Konami's arcade game Dance Dance Revolution and, later, with the Nintendo Balance Board for the Wii console. The integration of VR into computer exergames showed the potential to play a significant role in balance rehabilitation more than a decade ago [28]. The immersive and engaging nature of VR in fact positively impacts patient motivation and adherence to rehabilitation programs [1], by enabling therapists to adjust the intensity and difficulty of tasks to match the patient's abilities, so as to promoting gradual progress and improving treatment outcomes.

One review reported that patients often respond positively to VR balance rehabilitation, due to its engaging and enjoyable nature, sense of control, personalized approach and feedback mechanisms [3], with significant effects in cerebral palsy children [6] and chronic stroke patients [8]. This also holds true for the elderly population, with studies showing positive perceptions of VR use in this age group [5] and significant improvement in agility, balance and functional mobility in older women [27], holding the importance of assessing the movement challenge and preventing from engagement decay because of adaptation [9, 14]. However, training on specific exergames does not imply the transfer of acquired skills across other postural demands [23]. Furthermore, a recent review about the application of VR in balance training programs in clinical practice warns about the lack of enough accurate objective methods and test methodologies to evaluate the influence of VR on balance rehabilitation [19].

In this paper, we will present the collaborative design, development and preliminary testing of two VR games that aim to train the ability of patients within specific domains of balance. Section 2 describes the methods used for the game design, then provides a description of the gameplay also discussing the multisensory feedback, and finally presents the evaluation framework. Section 3 presents the results of an expert evaluation, as a summary of the feedback provided by physiotherapists with extensive experience in balance rehabilitation. In the end, Section 4 shortly discusses and concludes the work.

2 METHODS

Two VR scenarios were designed through a collaboration between researchers in interaction design, Unity3D developers, and a physiotherapist with extensive experience in balance disorders who participated in several brainstorming and workshops sessions organized at the Multisensory Experience Laboratory of Aalborg University in Copenhagen. As a starting point, the physiotherapist tested across approximately one hour a Meta Quest 3 headset and the related controllers, that were proposed by the research team as a currently optimal VR device in terms of usability, accuracy and application development time under the Unity3D programming environment. Balancing tasks eligible for the two respective VR experiences were selected from the STABLE (Specific Training According to BaLance Evaluation) rehabilitation protocol [20] adopted by the Hospital Herlev and Gentofte in Ballerup, Denmark, which is based on the Balance Evaluation System's Test (BESTest) [16]. STABLE divides the balance ability of a patient within six domains of interest: power, stability limits, anticipatory turning, reactive stepping, sensory orientation and cognitive-motor interaction, with the aim to measure individual balance abilities and to focus on exercises towards specific limitations and monitor change. By covering four such domains, the resulting applications hence form an initial

testbed of the use of STABLE in VR exergames.

2.1 Game A: Stability Limits Training

Stability limits are exercised in this exergame, by training the user's ability to extend and control their body's center of mass over a limited floor region. To this goal, game players are prompted to reach out and grab two different objects. Apples and oranges randomly spawn in the front, and then move slowly towards them. Players grab and then sort the fruits in the corresponding fruit crates, located at the opposite side of the grabbing hand (see Figure 1). The task requires they grab the oranges with their right hand and the apples with the left, per the association between fruits and controllers which share the same color in the scenario. This makes the players reach in several directions with both hands, with consequent shifts of their center of mass as they cannot trespass a floor region indicating where they must stand: if they move away from this region they are warned and guided back to it. The difficulty increases with score throughout the game, with objects spawning from more directions at increasing spawn rate. Successful sorting of an object leads to an increase of the score and to a new object spawning. To further incentivize players to reach as far as they can, points are awarded proportionally to how far the objects are grabbed.

One of the main benefits of VR is immersion in the experience [18]. To achieve this, coherence of the sensory feedback is crucial. In contrast, inconsistent input across sensory modalities can increase the cognitive load of the user during the VR experience [4]. Hence, a careful consideration of the accuracy and alignment of the sensory feedback in the virtual environment is necessary. A careful design of multisensory feedback in VR results in a positive effect on the user experience in more than 90% of the studies that were systematically reviewed by Apostolou and Liarokapis [2]. Therefore, we aimed to design a multisensory experience that includes audio and haptics that, while aiding patients in their task, also increases their sense of presence.

2.1.1 Audio

Auditory feedback can be effective in directing users' attention to a stimulus, with positive effects on reaction time [13]. Hence, it was chosen to attach interactive sounds to (or *sonify*) the objects that are grabbed during the task. Unity3D's Audio sources are associated to such objects, which are then spatialized via the Meta XR Audio SDK plugin. This way, even if an object is outside of the player's field of view, their attention can be guided towards it. Audio clips of sampled bell sounds are played in a loop, as such sound sources are traditionally associated to alerting meanwhile being pleasant [25]. Additionally, the use of ecological natural sounds, as is the case when using high quality samples of original recordings, is also recommended in virtual environments [17]. The two object types are associated to bell samples with different timbres, so that the player can quickly tell them apart. In order to distinguish between incoming objects of the same type, the pitch of each sound source is randomly adjusted by a factor between 0.9 and 1.1 during the creation of the respective object. In summary, the timbre describes the object type, while its pitch describes each individual object. To avoid cluttering of the soundscape, no more than two objects are sonified at any time. An object becomes silent when the player grabs and sorts it into the crate, or if the object disappears from the scene because the user did not grab it in time; at that point the oldest object in the scene starts to play, as it is closest to the user and thus requires the highest attention. Overall, the audio scene was purposefully kept minimal in order not to distract or overwhelm the user.

The feeling of immersion a user experiences in a VR game can be enhanced by consistent reactive sounds reflecting their dynamic character [21]. For this purpose, a "swoosh"-type sound was introduced when the user moves a grabbed object. Its loudness depends on the velocity of movement, creating a realistic and reactive au-

ditory feedback. This effect is generated by low-pass filtering a sample of white noise, with a medium-high resonance, at a cut-off frequency of 700 Hz.

In order to reinforce engagement, a “bubble pop” sound is provided when the user places an object in the correct crate. This sound presents a short attack and decay and a bright, crisp timbre. One last auditory event consists of a warning sound playing when the user walks outside of the allowed floor region between the two boxes. This event is enabled by attaching a sound source to the floor region itself, again spatialized using the Meta XR Audio SDK. Three different audio clips are used for the warning, consisting of sea waves breaking on a shore with increasing intensity. The farther a player moves away from the admitted floor region, the more intense the breaking waves are. Ecological soundscapes can be pleasant even if they are used for warning purposes, in which case they could conversely be disturbing [17].

2.1.2 Haptics

Benefits of coupling audio and haptic feedback in VR environments have been suggested by a number of studies [2, 12]. Haptic feedback has been shown to improve interaction, detection of stimuli and spatial awareness. For this purpose, gameplay interactions in Game A were augmented with haptic feedback using the Meta Haptics SDK for Unity. A subtle haptic response, designed with Meta Haptics Studio, was created from the same “bubble pop” audio clip used for audio feedback during the sorting task. This haptic effect is integrated into the Oculus controller to describe two events: when grabbing an object, to enhance interaction, and when placing it in the correct crate, to reinforce positive feedback for completing the task.

2.2 Game B: Sensory Integration Training

Training application for sensory integration and dynamic balance were exercised in this game, that challenges users to follow objects with their head. This type of rehabilitation is beneficial as it engages the visual, vestibular, and proprioceptive systems, promoting improvements in postural stability, gaze control, and coordination.

To address this need, a scenario was developed in which players interact with flying objects again represented with apples and oranges, that move toward them in a dynamic 3D environment. Players use the controller by pressing the trigger button to fire with a laser aimed in the direction they are looking, mimicking Superman when shooting lasers from his eyes. When hit, the fruits split into two pieces, providing immediate visual feedback (as shown in Figure 2). The game requires that players track flying objects with their head. This requirement asks for precise motor actions, effectively challenging their sensory integration and dynamic balance in an engaging and immersive environment.

The game difficulty is adjustable, allowing the physiotherapist to customize the experience for each session. Using sliders, the physiotherapist can modify parameters such as the spawn rate of the fruits, their speed as they move toward the user, and the spread of their spawn locations along the horizontal and vertical axes.

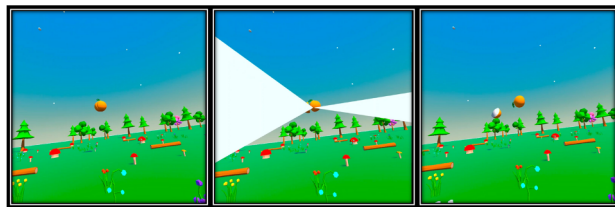


Figure 2: Illustration of the players' viewpoint when spotting the fruit and shooting it with the laser coming out of their eyes

2.3 Preliminary evaluation

The exergames were preliminary evaluated at the Department of Occupational and Physical Therapy of the University Hospital Herlev and Gentofte. Five physiotherapists and one medical doctor with extensive experience in working with individuals with balance disorders participated in the evaluation. Five of them had not participated to the initial design workshops. These professionals, who primarily work with older adults, tested the application and provided think-aloud feedback during their sessions.

Following individual game sessions, an evaluation workshop was conducted where all six health professionals collectively discussed their experiences and suggested improvements. This collaborative session facilitated deeper insight into the usability, effectiveness, and potential areas for improvement of the application. Their feedback forms the basis for refining the application to meet the therapeutic needs of people with balance challenges.

3 RESULTS

3.1 Game A: Stability Limits Training

The physiotherapists expressed that the game effectively makes the player reach and move their body's center of mass over their base of support in an engaging way.

The mechanic of the fruits moving toward the player was discussed. As the fruit moves close to the player it becomes easier to reach, hence progressively eliminating the need to move your center of mass in order to complete the grabbing task. A suggestion for circumventing this was for the fruits to be stationary, and then spawn at different distances to the player, ensuring that the player has to move their center of mass.

It was also emphasized that it was important to be able to start the game at different difficulties according to the patient's abilities and needs, as some might not be able to reach very far in the beginning of their training. Finally, it was asked to empower the auditory feedback concerning the display of off-task information such as the score level, for two reasons: to minimize visual distraction during the sorting task, as well as to boost engagement by displaying score changes through corresponding auditory arousal, for instance by increasing their pitch as score progresses.

3.2 Game B: Sensory Integration Training

The physiotherapists unanimously agreed that following objects with the head and shooting lasers to hit the fruits would effectively motivate their users to perform movements essential for sensory integration training. They highlighted that such tasks engage the visual, vestibular, and proprioceptive systems, providing a dynamic and interactive method to improve coordination and postural control. The shooting mechanic and audio-visual feedback were described as enjoyable and engaging, enhancing users' motivation to train regularly.

Physiotherapists particularly appreciated the ability to customize the difficulty of the game through adjustable parameters such as the spawn rate, the speed of the object, and the spread of the movement. They noted that this flexibility is essential, as their patients have varying abilities and therapeutic needs. Initially, some presets for the difficulty metrics organized in increasing game levels can be used, to set a performance baseline for each patient. They also emphasized that the game is suitable for home-based training and structured sessions at the hospital.

Several suggestions for improvement were provided. They fostered tools to monitor and receive feedback on user performance, which could help tailor the therapy further and track progress over time. Additionally, they suggested introducing a feature allowing users to turn 180 degrees to shoot objects from behind, which could benefit patients requiring more advanced balance and rotational movement training. Finally, they recommended implementing environmental variability, such as changing the virtual environment

after completing a certain number of tasks, to keep the experience fresh and prevent monotony during extended sessions.

4 DISCUSSION AND FUTURE WORK

The two VR exergames so far realize only a portion of the STABLE protocol. More tasks exportable to VR are in due realization to complete the protocol. Obviously the acceptability of the resulting application from the target users' group is left to a future step of this collaborative project. Such a step is planned to be carried out within a timeframe of several weeks. In particular, it will be essential to prime all players with balance disorders by asking them to just wear the headset and observe the real surroundings using its see-through feature: only if feeling comfortable in this condition could they proceed to assess its potential as a rehabilitation tool.

The design, development, and evaluation process carried out so far has not only provided the foundation for the remaining activities but also provided a key insight: prior to any user test, it is preferable to involve independent expert groups respectively during the design and preliminary evaluation. In our case, the independence on the design step of five out of six health professionals provided an especially constructive criticism of the VR exergames during the subsequent evaluation workshop participated by them. The feedback provided by these experts was highly valuable for the design stage. In terms of sample size with respect to the number of experts providing feedback to the game design, five individuals with varying experiences and expertise can greatly broaden the perspective of the design team.

The effectiveness of the games as a rehabilitation tool will be assessed based on the feedback of the patients and the objective evolution of their balance performance with respect to the balance domains of interest, evaluated using the standard tests of the STABLE protocol.

Future work also includes the implementation of a dynamic difficulty adjustment methodology with the scope of optimizing the gameplay experience and avoiding the risk of monotony. This can be ensured by providing a difficulty level that keeps the player engaged, tailored to each individual. One potential method is to fit a best-estimate difficulty/performance curve for each user, from which informed decisions of difficulty metrics can be inferred. Tracking such a curve at various stages throughout rehabilitation and analyzing its progression can also be used to monitor the patient's performance over time.

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